Conformal covariance of the supercritical Liouville quantum gravity metric

October 25, 2026

Abstract

We give a domain-field formulation of conformal covariance for the supercritical Liouville quantum gravity metric. For a zero-boundary Gaussian free field plus a possibly dependent continuous function on a connected Greenian domain, the metric has a measurable representative depending only on the field. Under a fixed conformal change of coordinates, the metrics agree almost surely at every pair of points, including singular points. Representatives can be chosen for any prescribed countable family of field laws and maps. We organize the result in the form of Gwynne–Miller's conformal-covariance theorem while retaining the quantifiers of the accompanying Lean statement. The proof derives this formulation from realization-based covariance and realization independence, with an explicit construction of domain metrics and descent to the original probability space.

1 Introduction

Fix $\xi > \xi_{\mathrm{crit}}$ and let $Q = Q(\xi) \in (0, 2)$ be the LFPP distance exponent. Let $H \mapsto D_H$ be a strong whole-plane LQG metric with parameter ξ and one fixed normalization. Here *strong* refers to Ding–Gwynne's axioms [1, Definition 1.7]: the length-space property, locality, continuous Weyl scaling, and translation and dilation covariance. Their existence and uniqueness theorem identifies this assignment up to a deterministic positive factor. We use the same assignment throughout.

Distances may take the value $+\infty$. A domain means a nonempty connected open subset of $\mathbb{C}$. The main results concern Greenian domains, on which the zero-boundary GFF is defined, without any boundary-regularity assumption. Write $\mathcal{D}'(U)$ for distributions on U with the measurable structure generated by test-function evaluations. Set $\mathcal{M}(U) = [0, \infty]^{U \times U}$ with its product measurable structure. The representatives below are measurable maps into $\mathcal{M}(U)$; their values are required to be metrics almost surely under the specified field laws, not at every distribution.

Definition 1.1 (Domain fields)**.** An admissible field on a Greenian domain U is a measurable random distribution h on $(\Omega, \mathcal{F}, \mathbb{P})$ with

$$h = h_0^U + f \quad \text{almost surely,}$$

where h_0^U is a zero-boundary GFF on U and f is a measurable random element of $C(U, \mathbb{R})$. The decomposition is supplied on the given probability space. The function f may depend on h_0^U, need not be bounded, and need not extend continuously to ∂U.

A whole-plane realization of h is a pair (H, F) with H a whole-plane GFF plus a continuous function and F a measurable random element of $C(U, \mathbb{R})$ such that $h = H|_U + F$ almost surely.

Realizations are also allowed on measure-preserving extensions $\pi : \Omega' \to \Omega$, with $h \circ \pi = H|_U + F$. For a realization define

$$\mathsf{D}^U_{H,F}(x,y) = \inf_{P:x\to y,\ P\subset U} \int_0^{\mathrm{len}(P;D_H)} e^{\xi F(P(t))}\,\mathrm{d}t. \tag{1}$$

Paths are rectifiable and parametrized by D_H-length. An empty infimum is $+\infty$, and a constant path has length zero. This is an *internal* distance in U.

Definition 1.2 (A version for a specified field)**.** A measurable $\Phi_U : \mathcal{D}'(U) \to \mathcal{M}(U)$ is a metric version for h if, on every measure-preserving extension carrying a realization (H, F) of $h \circ \pi$,

$$\Phi_U(h \circ \pi) = \mathsf{D}^U_{H,F} \quad \text{almost surely as functions on } U^2. \tag{2}$$

Whenever such a version is asserted, existence of a realization is included. The null event may depend on the extension and realization.

For a deterministic conformal bijection $\phi : U \to V$, write

$$h^\phi = h \circ \phi^{-1} + Q \log |(\phi^{-1})'|. \tag{3}$$

Composition of distributions uses pullback of test functions with the appropriate Jacobian.

Theorem 1.3 (Conformal covariance)**.** *Let U be a connected Greenian domain, let $\phi : U \to V$ be a fixed deterministic conformal bijection, and let h be an admissible field on U. There exist measurable maps*

$$\Phi_U : \mathcal{D}'(U) \to \mathcal{M}(U), \qquad \Phi_V : \mathcal{D}'(V) \to \mathcal{M}(V)$$

which are metric versions for h and h^ϕ, respectively. Both fields have whole-plane realizations on product extensions of their original probability space. Almost surely on $(\Omega, \mathcal{F}, \mathbb{P})$,

$$\Phi_U(h)(x,y) = \Phi_V(h^\phi)(\phi(x), \phi(y)) \qquad \textit{for all } x, y \in U. \tag{4}$$

Thus, with $D^U_h = \Phi_U(h)$ and $D^V_{h^\phi} = \Phi_V(h^\phi)$,

$$D^U_h(x,y) = D^V_{h^\phi}(\phi(x), \phi(y)).$$

This is the forward convention of Gwynne–Miller's Axiom IV and Theorem 1.3 [2]. The version and existence clauses identify the metrics with the fixed whole-plane construction and exclude vacuous choices of representatives. Equality is simultaneous in the endpoints, including singular endpoints, with exact constant one. The event may depend on the fixed map, parameter, and field law.

Theorem 1.4 (A prescribed countable family)**.** *Keep ξ, Q, D, U, V fixed. For each index i in a countable set I, let $\phi_i : U \to V$ be a deterministic conformal bijection and let h_i be admissible on $(\Omega_i, \mathcal{F}_i, \mathbb{P}_i)$. One pair Φ_U, Φ_V can be chosen before the index i so that, for every i, both product realizations exist, the maps are versions for h_i and $h_i^{\phi_i}$ on every realization extension, and*

$$\mathbb{P}_i\Big[\Phi_U(h_i)(x,y) = \Phi_V(h_i^{\phi_i})(\phi_i(x), \phi_i(y)) \textit{ for all } x, y \in U\Big] = 1.$$

The representatives are chosen after the law, or prescribed countable family of laws, is specified. We do not assert one Φ_U chosen before *all* admissible laws, or a common sample event for an uncountable collection of conformal maps.

Proof organization and sources

We construct the domain metric on a realization, represent it measurably in terms of the field, and then transfer realization-based covariance to the original probability space. A positive-weight mixture supplies common representatives for a countable family.

The analytic covariance input is the released Lean theorem [4], stated precisely in Proposition 3.1. This rewrite proves the domain-field formulation from that theorem; it does not reproduce its entire analytic comparison proof as an independent proof. Section 3 explains that proof's structure and gives a repaired endpoint estimate. Appendix A records the declaration correspondence.

Gwynne–Miller's theorem supplies the form of the statement, not the supercritical input. Devlin's simultaneous coordinate-change theorem [3, Theorem 1.1] assumes $\xi < \xi_{\mathrm{crit}}$ and is not used here.

2 Domain metrics and realizations

Lemma 2.1. *An admissible field on U has a whole-plane realization on a measure-preserving product extension $\Omega \times \mathcal{D}'(\mathbb{C}) \to \Omega$. The transformed field h^ϕ is admissible on V and has such a realization as well.*

Proof. The whole-plane Markov decomposition on U gives a reference coupling

$$H|_U = Z + \mathfrak{h}_U,$$

where Z has the zero-boundary GFF law and $\mathfrak{h}_U$ has a continuous harmonic version on U. Couple the reference whole-plane field to the given h_0^U using its conditional law given Z. This keeps the full joint law of (h, h_0^U, f) unchanged. On the product extension,

$$h \circ \pi = H|_U + (f \circ \pi - \mathfrak{h}_U).$$

The correction is continuous. No independence between f and h_0^U is required.

Conformal invariance of the zero-boundary GFF gives

$$h^\phi = (h_0^U \circ \phi^{-1}) + (f \circ \phi^{-1} + Q \log |(\phi^{-1})'|).$$

The first summand is a zero-boundary GFF on V and the second is a measurable random continuous function. The image V is connected and Greenian. Apply the same realization construction on V. □

The corresponding product construction is `exists_prod_realization_of_zeroBoundary`. It includes the requisite measurable continuous versions.

2.1 The domain forms of the metric axioms

Formula (1) is the intrinsic length metric obtained by weighting D_H-length. For every open $W \subset U$ it satisfies

$$\mathsf{D}^W_{H,F|_W}(x,y) = \mathsf{D}^U_{H,F}(x,y;W). \tag{5}$$

For $g \in C(U,\mathbb{R})$ it satisfies

$$\mathsf{D}^U_{H,F+g} = e^{\xi g} \cdot \mathsf{D}^U_{H,F}, \tag{6}$$

where the right-hand side is path-length reweighting internal to U. These identities hold on the realization-level sample event simultaneously in continuous corrections.

To justify functions unbounded near ∂U, observe that every path used in a distance has compact image inside U. Continuous weights are bounded above and below on that image. The weighted length formula applies there, successive reweighting multiplies the weights, and infimizing over paths gives (5)–(6). If F extends to $\widehat{F} \in C(\mathbb{C})$, then

$$\mathsf{D}^{U}_{H,F} = D_{H+\widehat{F}}(\cdot,\cdot;U).$$

Otherwise this identity is applied only on relatively compact subsets using cutoff extensions.

Proposition 2.2 (Realization independence)**.** *If (H,F) and $(\widehat{H},\widehat{F})$ realize the same domain field on one probability space, then*

$$\mathsf{D}^{U}_{H,F} = \mathsf{D}^{U}_{\widehat{H},\widehat{F}} \quad \textit{almost surely on } U^2.$$

This statement applies on any open set for which the realizations are supplied.

This is the released theorem `domainMetric_ae_eq_of_isDomainRealization`. It is a pathwise identification on the coupling, not merely equality in distribution. Its proof uses locality for cutoff whole-plane fields on a countable compact exhaustion, followed by the internal path construction. In particular, fields agreeing on an open overlap have equal internal metrics on that overlap. The domain length, restriction, locality, and Weyl statements are developed in `Domain/Axioms.lean`. They are the domain counterparts of the first three axioms in Gwynne–Miller's presentation. They do not assert a single measurable representative valid before every field law is specified.

3 The analytic covariance theorem

Proposition 3.1 (Covariance for realized fields)**.** *Let $\xi > \xi_{\mathrm{crit}}$, $Q = Q(\xi)$, and D be any strong whole-plane LQG metric with these parameters. Let U, V be connected Greenian domains and $\psi : U \to V$ a fixed deterministic conformal bijection. If k on V has a realization (H,F), the field $k\circ\psi + Q\log|\psi'|$ has a realization on a measure-preserving extension. On every such extension and for every such realization (G,K), almost surely,*

$$\mathsf{D}^{V}_{H\circ\pi,F\circ\pi}(\psi(u),\psi(v)) = \mathsf{D}^{U}_{G,K}(u,v) \quad \textit{for all } u,v \in U. \tag{7}$$

We use the frozen `conformal_covariance` theorem of [4]. The existence clause is part of the theorem. It imposes no additional connector or comparison assumption on a strong metric.

Here is the structure of its analytic proof. First compare the source metric d with the conformally pulled-back metric e. Annular estimates and a filled-set loop chain give a deterministic finite bi-Lipschitz constant. The filled-set construction ensures connectivity and includes the terminal loop's cost. Small conformal maps are then compared with affine tangents, using a smooth field normalization, compact confinement, and lower semicontinuous metric convergence. For any sequence of fields, a countable mixture supplies one locality functional valid for all its laws. The confinement event uses a strict annular barrier margin.

Recovery in the lower semicontinuous topology moves endpoints. The following deterministic estimate explains how short loops recover the actual endpoints, with the radial cost fully accounted for.

Lemma 3.2 (Two-scale endpoint recovery)**.** *Suppose d_n, e_n are extended length metrics on domains eventually containing every compact subset of $\mathbb{C}$. Their finite-length paths are Euclidean continuous, $C^{-1}d_n \le e_n \le Cd_n$, and both sequences converge locally in the lower semicontinuous topology to the same extended metric d.*

Let $u_n \to u$, $v_n \to v$ with $u \neq v$, and suppose $L_n = d_n(u_n, v_n) \to L < \infty$. Suppose $s_j \downarrow 0$ and, for $x = u_n, v_n$, every fixed j, and all sufficiently large n,

$$d_n(x, \partial B_{s_j}(x)) \le M s_j^\theta, \qquad d_n(\text{around } A_{s_j,\sqrt{s_j}}(x)) \le M s_j^\theta \tag{8}$$

for $0 < M < \infty$ and $\theta > 0$. Then

$$\limsup_n e_n(u_n, v_n) \le d(u, v) \le L.$$

Proof. The lower-bound clause of convergence gives $d(u, v) \le L$. Choose recovery endpoints $a_n \to u$, $b_n \to v$ for e_n and paths Q_n between them whose e_n-lengths tend to $d(u, v)$. Fix k large enough that the s_k-balls about the two limiting endpoints are disjoint. Choose j with $\sqrt{s_j} < s_k$.

For large n, the recovery endpoints lie inside the corresponding s_j-balls. Use the first bound in (8) at scale s_k to reach $\partial B_{s_k}(u_n)$ at d_n-cost at most $2Ms_k^\theta$. Use the second bound at scale s_j to find a surrounding loop of length at most $2Ms_j^\theta$. The radial path crosses the loop since $s_k > \sqrt{s_j}$. Following the radial path to its first intersection and then a portion of the loop connects u_n to any prescribed point of the loop at cost at most $2M(s_k^\theta + s_j^\theta)$. Repeat at v_n.

Pointwise metric comparison compares path lengths by taking suprema over partitions. The path Q_n crosses both loops by Euclidean continuity. Splicing its middle part to the connectors gives

$$\limsup_n e_n(u_n, v_n) \le d(u, v) + 4CM(s_k^\theta + s_j^\theta).$$

Choose $j = j(k) \to \infty$ as $k \to \infty$ and let $k \to \infty$. □

A short path only to radius s_j need not reach a loop in $A_{s_j,\sqrt{s_j}}$. The larger-radius estimate is therefore essential. This repairs the connector step of the earlier manuscript.

The analytic argument next builds good annuli whose regular geodesic crossings have comparison coefficient $1 + \eta$. Near-independence across annular scales and a finite lattice cover give an event valid for all relevant geodesics. A definite fraction of each retained interval lies in a good crossing. If C_* is the optimal upper comparison constant, this yields

$$C_{\text{new}} = (1 + \eta) + \frac{A}{A + 1}(C_* - 1 - \eta) < C_*, \qquad 0 < \eta < C_* - 1.$$

The retained intervals are selected using a last-exit/first-entry corridor so that the geodesic endpoints lie outside the enlarged balls. The contradiction gives $C_* = 1$; the inverse-map argument gives equality. The complete analytic proof, including the probability estimates, is in the `BiLip`, `Tangent`, `Recovery`, `Cover`, `Improve`, and `Root` modules of [4]. This summary is not a replacement for those proofs.

4 Measurable metric versions

Lemma 4.1 (A representative for one realization)**.** *For a specified field h with a realization (H, F) on U, there is a measurable $\Phi_U : \mathcal{D}'(U) \to \mathcal{M}(U)$ such that $\Phi_U(h) = \mathsf{D}^U_{H,F}$ almost surely as functions on U^2.*

Proof. Choose an increasing relatively compact open exhaustion $W_n \Subset U$ which eventually contains every compact subset of U. Choose a measurable cutoff extension $F_n \in C(\mathbb{C})$ equal to F on W_n: multiply by a deterministic continuous cutoff compactly supported in U, and extend by zero.

Locality of the whole-plane metric gives a measurable $A_n : \mathcal{D}'(W_n) \to \mathcal{M}(W_n)$ representing $D_{H+F_n}(\cdot,\cdot;W_n)$ as a function of $(H+F_n)|_{W_n} = h|_{W_n}$. This representative is chosen for the law under consideration. Define

$$\Phi_U(k)(x,y) = \inf_{\substack{n\geq 1 \\ x,y\in W_n}} A_n(k|_{W_n})(x,y). \tag{9}$$

For each pair this is a countable infimum of measurable functions, so the map into the evaluation-product space $\mathcal{M}(U)$ is measurable.

Intersect the countably many locality and Weyl events. On this event, each term is the internal weighted distance on W_n, simultaneously for all its endpoint pairs. The infimum equals (1), since every path in U has compact image and is contained in some W_n. The resulting equality holds as a function on all of U^2. □

This is `exists_measurable_domainMetric`. In particular, no unjustified factorization into an uncountable product is needed.

Lemma 4.2 (Descent through a product extension)**.** *Let μ be a probability measure on $\Omega \times X$ with first marginal $\mathbb{P}$, where X is a nonempty standard Borel space. If a predicate $\mathcal{A}(\omega)$ holds for μ-almost every (ω, x), then it holds for $\mathbb{P}$-almost every ω, with almost-everywhere statements understood in the completed measure.*

Proof. Choose a measurable μ-null set N containing all pairs where $\mathcal{A}(\omega)$ fails. Disintegrate μ over its first marginal with conditional probabilities $K(\omega, \mathrm{d}x)$. For almost every ω, $K(\omega, N_\omega) = 0$. If $\mathcal{A}(\omega)$ fails, $N_\omega = X$ and this probability equals one. Thus the failing set lies in a measurable $\mathbb{P}$-null set. □

This is `ae_of_ae_fst`. It applies to the predicate asserting equality at every endpoint pair, without first assuming measurability of that uncountable intersection.

Lemma 4.3 (Every realization extension)**.** *If a measurable Φ_U agrees with the metric on one product realization of h, then it agrees with the metric on every realization on every measure-preserving extension of the same base field.*

Proof. Disintegrate the fixed product realization over Ω. Given another realization on $\pi : \Omega' \to \Omega$, sample the auxiliary whole-plane field by this conditional law at $\pi(\omega')$. The resulting common refinement on $\Omega' \times \mathcal{D}'(\mathbb{C})$ projects with the correct laws to both realizations.

On the refinement, pull back the equality defining Φ_U. Proposition 2.2 identifies its realized metric with the competing realized metric. The desired equality therefore holds almost surely on the refinement. It depends only on ω', so Lemma 4.2 descends it to Ω'. □

The original probability space need not be standard Borel. The auxiliary distribution space supplies the conditional kernel. These constructions are `product_common_refinement` and `version_on_all_extensions`.

5 Proof of Theorem 1.3

By Lemma 2.1, choose product realizations of h and h^ϕ over their original probability space. Lemma 4.1 supplies measurable representatives Φ_U, Φ_V. Lemma 4.3 makes them versions on every realization extension, as required by Definition 1.2.

Apply Proposition 3.1 with the domains interchanged and with $\phi^{-1} : V \to U$. The field on its target is h; the pulled-back field on its source is $h \circ \phi^{-1} + Q \log |(\phi^{-1})'| = h^\phi$. On a further product extension carrying the target realization,

$$\mathsf{D}^U_{H,F}(x, y) = \mathsf{D}^V_{G,K}(\phi(x), \phi(y)) \quad \text{for all } x, y \in U$$

almost surely. All fields are understood to be pulled to the common extension. The version clauses replace these two realized metrics by $\Phi_U(h)$ and $\Phi_V(h^\phi)$.

The resulting all-endpoints identity depends only on the original sample ω. Apply Lemma 4.2 through the two product extensions to obtain (4) on the original probability space. This proves Theorem 1.3.

Every equality used here takes values in $[0, \infty]$; no infinite distance is subtracted or divided. Singular endpoints are therefore retained. The whole-plane assignment and its normalization are fixed on both sides, so there is no extra multiplicative constant.

6 Countably many laws and maps

Lemma 6.1. *For any prescribed countable family of realized fields $h_i = H_i|_U + F_i$, one measurable map Φ_U represents all their domain metrics almost surely under their respective laws.*

Proof. For a nonempty family choose positive weights w_i summing to one and form the disjoint-union law $\sum_i w_i \mathbb{P}_i$. On component i set $(h, H, F) = (h_i, H_i, F_i)$. Normalize the GFF component of each H_i in the same way, absorbing any additive constant into its continuous correction. The mixture GFF component has this same law, while the continuous correction may depend on both the field and the index. Thus the mixture is again a whole-plane realization.

Apply Lemma 4.1 once to this mixture. Every component has positive weight, so its almost-sure identity holds under every $\mathbb{P}_i$, with the same Φ_U. An empty family imposes no condition. □

Proof of Theorem 1.4. Construct product realizations for h_i and $h_i^{\phi_i}$ over each Ω_i. Apply Lemma 6.1 separately to the source and target families to choose one pair Φ_U, Φ_V. For every i, Lemma 4.3 makes these representatives versions on every realization extension. The proof of Theorem 1.3, with the fixed pair of representatives, now applies to h_i and ϕ_i and gives equality on Ω_i. □

The mixture selects functions; it does not replace the original probability spaces by a single asserted sample space. If a countable family does share one probability space, its full-probability events can be intersected. A countable-mixture construction does not establish one representative for every admissible law.

A Correspondence and scope

This rewrite uses public revision `de92cf632a47` of the accompanying Lean repository. Paths below are relative to `SupercriticalConformalCovariance`.

Manuscript	Released declaration or module
Theorem 1.3	`GMForm.gm_axiomIV_on_base`
Theorem 1.4	`GMForm.exists_countably_universal_gm_axiomIV_on_base`
Proposition 3.1	`conformal_covariance`
Proposition 2.2	`domainMetric_ae_eq_of_isDomainRealization`
Domain construction	`Frozen/DomainMetric.lean`; `Domain/Axioms.lean`
Product realization	`GMForm.exists_prod_realization_of_zeroBoundary`
Versions on extensions	`GMForm.version_on_all_extensions`
Base-space descent	`GMForm.ae_of_ae_fst`
Countable representative	`GMForm.exists_countably_universal_domain_metric`

The main theorems retain the Lean statements' connected Greenian domains, supplied zero-boundary decomposition, measurable random continuous correction, fixed parameter, fixed normalization, and versions on every realization extension. Their equality is on the field's original probability space and is simultaneous in all endpoints.

The frozen covariance and realization-independence theorems are the analytic inputs. The domain-field theorems are supporting declarations derived from them. The release guide distinguishes their audit status from the frozen theorem seals. This prose rewrite is not a new kernel build. The separate `Root.conformal_covariance_all_xi` theorem removes the supercritical restriction for the realization-based statement. We keep $\xi > \xi_{\mathrm{crit}}$ in Theorems 1.3 and 1.4, matching their `GMForm` declarations.

The earlier manuscript is superseded by this domain-field formulation. Its radial connector is repaired in Lemma 3.2; the other analytic repairs are supplied by the released covariance theorem. The law-specific and countable-family scope remains distinct from one representative chosen before all admissible laws or all conformal maps.

References

[1] J. Ding and E. Gwynne. *Uniqueness of the critical and supercritical Liouville quantum gravity metrics*. Proc. London Math. Soc. 126 (2023), 216–333. https://arxiv.org/abs/2110.00177.

[2] E. Gwynne and J. Miller. *Conformal covariance of the Liouville quantum gravity metric for* $\gamma \in (0, 2)$. Ann. Inst. H. Poincaré Probab. Statist. 57 (2021), 1016–1031. https://arxiv.org/abs/1905.00384.

[3] C. Devlin VI. *The coordinate change formula for the Liouville quantum gravity metric holds for all conformal maps simultaneously*. 2026. https://arxiv.org/abs/2603.04640.

[4] *Conformal covariance of the supercritical Liouville quantum gravity metric*. Accompanying Lean development and statement guide, public revision `de92cf632a47` (2026). Source and statement guide at the cited revision.